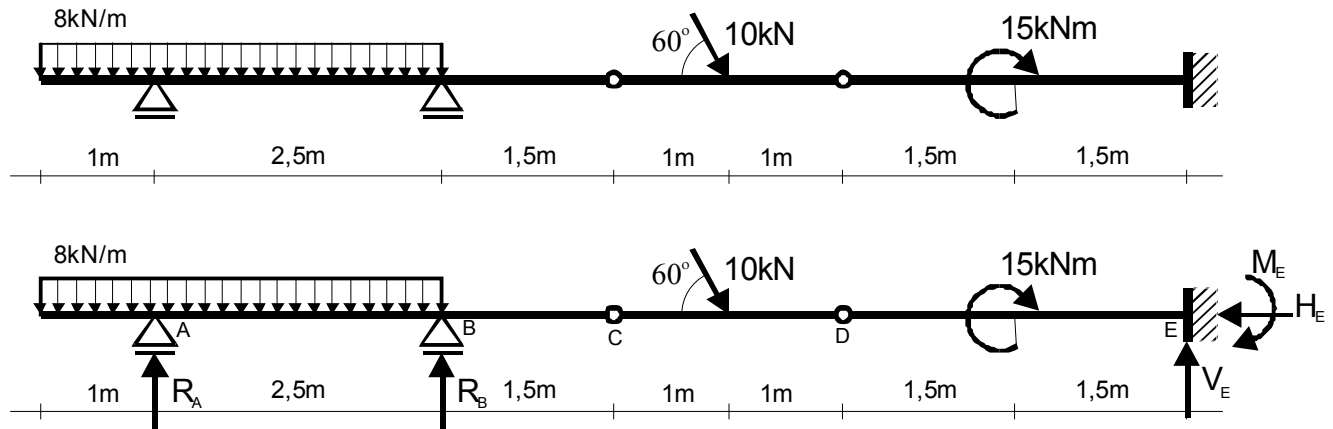


Wyznaczyć reakcje oraz siły wewnętrzne.

Reakcje.



$$\sum X : H_A - 10 \cos 60^\circ = 0$$

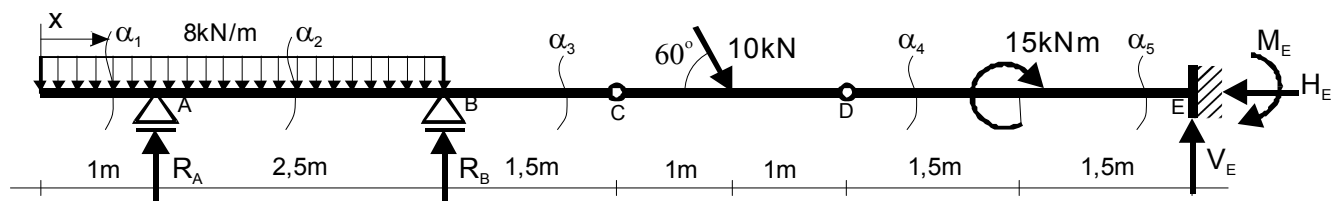
$$\sum Y : R_A + R_B + V_E - 8 \text{ kN/m} \cdot 3,5 \text{ m} - 10 \text{ kN} \sin 60^\circ = 0$$

$$\sum M_E : R_A \cdot 9 \text{ m} + R_B \cdot 6,5 \text{ m} + M_E + 15 \text{ kNm} - 10 \text{ kN} \sin 60^\circ \cdot 4 \text{ m} - 8 \cdot 3,5 \text{ m} \cdot \left(\frac{1}{2} 3,5 \text{ m} + 6,5 \text{ m} \right) = 0$$

$$\sum M_C^l : R_A \cdot 4 \text{ m} + R_B \cdot 1,5 \text{ m} - 8 \text{ kN/m} \cdot 3,5 \text{ m} \cdot \left(\frac{1}{2} 3,5 \text{ m} + 1,5 \text{ m} \right) = 0$$

$$\sum M_E^l : R_A \cdot 6 \text{ m} + R_B \cdot 3,5 \text{ m} - 8 \text{ kN/m} \cdot 3,5 \text{ m} \cdot \left(\frac{1}{2} 3,5 \text{ m} + 3,5 \text{ m} \right) - 10 \text{ kN} \sin 60^\circ \cdot 1 \text{ m} = 0$$

Oznaczenie przekrojów.



$$R_A = 17,002 \text{ kN}$$

$$\alpha_1 - \alpha_1 \quad x \in \langle 0; 1 \text{ m} \rangle$$

$$R_B = 15,328 \text{ kN}$$

$$\alpha_2 - \alpha_2 \quad x \in \langle 1 \text{ m}; 3,5 \text{ m} \rangle$$

$$H_E = 5 \text{ kN}$$

$$\alpha_3 - \alpha_3 \quad x \in \langle 3,5 \text{ m}; 6 \text{ m} \rangle$$

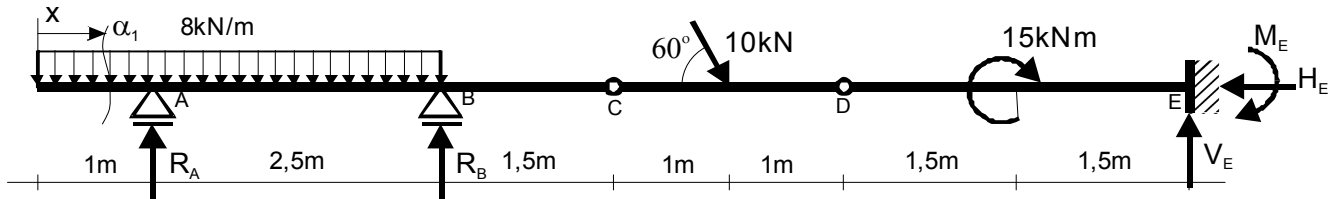
$$V_E = 4,330 \text{ kN}$$

$$\alpha_4 - \alpha_4 \quad x \in \langle 6 \text{ m}; 8,5 \text{ m} \rangle$$

$$M_E = -2,010 \text{ kN}$$

$$\alpha_5 - \alpha_5 \quad x \in \langle 8,5 \text{ m}; 10 \text{ m} \rangle$$

Przekrój α_1 .

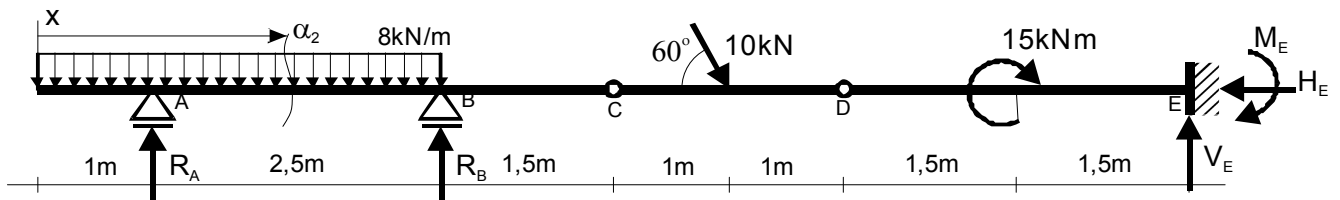


$$N_{\alpha_1} = 0$$

$$T_{\alpha_1} = -8 \text{ kN/m} \cdot x \quad \left| \begin{array}{l} x=0 \quad T_{\alpha_1} = 0 \\ x=1 \text{ m} \quad T_{\alpha_1} = -8 \text{ kN} \end{array} \right.$$

$$M_{\alpha_1} = -8 \text{ kN/m} \cdot x \cdot \frac{x}{2} = -4 \text{ kN/m} \cdot x^2 \quad \left| \begin{array}{l} x=0 \quad M_{\alpha_1} = 0 \\ x=1 \text{ m} \quad M_{\alpha_1} = -4 \text{ kNm} \end{array} \right.$$

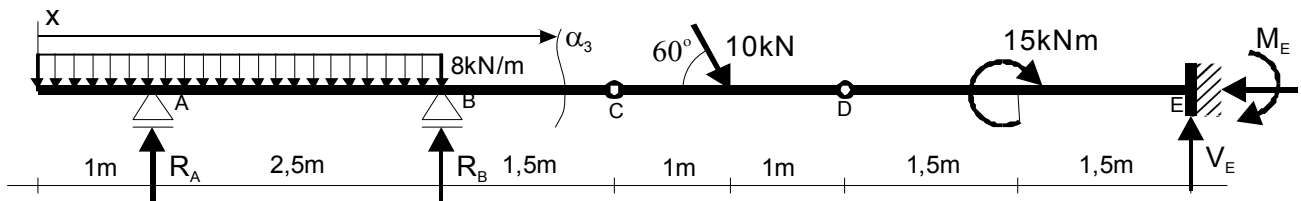
Przekrój α_2 .



$$N_{\alpha_2} = 0$$

$$T_{\alpha_2} = -8 \text{ kN/m} \cdot x + 17,002 \text{ kN} \quad \left| \begin{array}{l} x=1 \text{ m} \quad T_{\alpha_2} = 9,002 \text{ kN} \\ x=3,5 \text{ m} \quad T_{\alpha_2} = -10,998 \text{ kN} \end{array} \right.$$

$$M_{\alpha_2} = -8 \text{ kN/m} \cdot x \cdot \frac{x}{2} + 17,002 \text{ kN} \cdot (x-1 \text{ m}) = -4 \text{ kN/m} \cdot x^2 + 17,002 \text{ kN} \cdot x - 17,002 \text{ kNm} \quad \left| \begin{array}{l} x=1 \text{ m} \quad M_{\alpha_2} = -4 \text{ kNm} \\ x=3,5 \text{ m} \quad M_{\alpha_2} = -6,495 \text{ kNm} \end{array} \right.$$



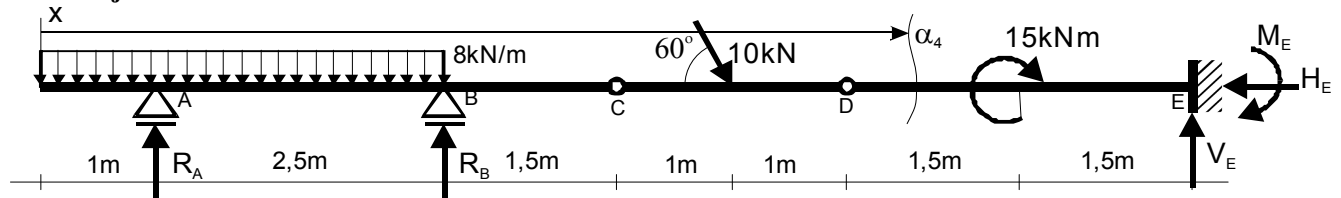
Przekrój α_3 .

$$N_{\alpha_3} = 0$$

$$T_{\alpha_3} = -8 \text{ kN/m} \cdot 3,5 \text{ m} + 17,002 \text{ kN} + 15,328 \text{ kN} = 4,33 \text{ kN}$$

$$M_{\alpha_3} = -8 \text{ kN/m} \cdot 3,5 \text{ m} \cdot \left(x - \frac{3,5 \text{ m}}{2} \right) + 17,002 \text{ kN} \cdot (x-1 \text{ m}) + 15,328 \text{ kN} \cdot (x-3,5 \text{ m}) = 4,33 \text{ kN} \cdot x - 21,65 \text{ kNm} \quad \left| \begin{array}{l} x=3,5 \text{ m} \quad M_{\alpha_3} = -6,495 \text{ kNm} \\ x=5 \text{ m} \quad M_{\alpha_3} = 0 \\ x=6 \text{ m} \quad M_{\alpha_3} = 4,330 \text{ kNm} \end{array} \right.$$

Przekrój α_4 .



$$N_{\alpha_4} = -10kN \cos 60^\circ = -5kN$$

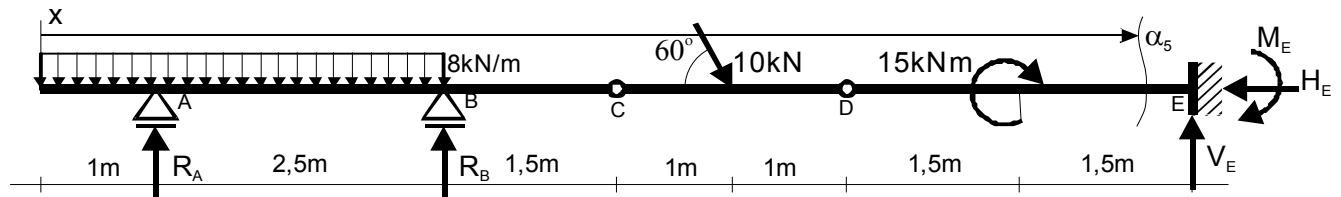
$$T_{\alpha_4} = 4,33kN - 10kN \sin 60^\circ = -4,33kN$$

$$M_{\alpha_4} = 4,33kN \cdot x - 21,65kNm - 10kN \sin 60^\circ \cdot (x - 6m) =$$

$$= -4,33kN \cdot x + 30,31kNm$$

$$\left| \begin{array}{ll} x = 6m & M_{\alpha_4} = 4,33m \\ x = 7m & M_{\alpha_4} = 0 \\ x = 8,5m & M_{\alpha_4} = -6,495kNm \end{array} \right.$$

Przekrój α_5 .



$$N_{\alpha_5} = -5kN$$

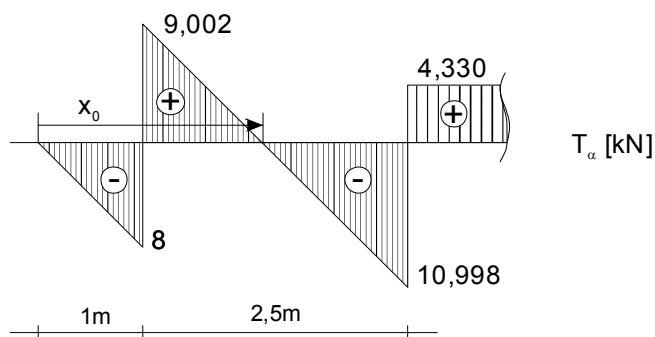
$$T_{\alpha_5} = -4,33kN$$

$$M_{\alpha_5} = -4,33kN \cdot x + 30,31kNm + 15kNm =$$

$$= -4,33kN \cdot x + 45,31kNm$$

$$\left| \begin{array}{ll} x = 8,5m & M_{\alpha_5} = 8,505kNm \\ x = 10m & M_{\alpha_5} = 2,01kNm \end{array} \right.$$

Obliczenie ekstremum momentów.



$$\frac{9,002kNm + 10,998kNm}{2,5m} = \frac{9,002kNm}{x_0 - 1m}$$

$$T_{\alpha_2} = -8kN / m \cdot x + 17,002kN = 0 \quad x_0 = 2,125m$$

$$M_{\alpha_2} = -4kN / m \cdot x^2 + 17,002kN \cdot x - 17,002kNm$$

$$M_{\alpha_2}(x_0 = 2,125m) = 1,065kNm$$

Wykresy sił wewnętrznych.

